Lesson plan · for the teacher

Ask a question, then collect

Data engineering · Level 1 · Lesson 2

Time	Materials
40 min	Board and chalk; Paper for tally sheets; Optional: a phone or computer with a spreadsheet app

Learning goals

- Turn a question into the data we need.
- Plan a tidy collection sheet: columns, units and codes.
- Collect with tally marks, and save the table as CSV.

Activity: Our class survey

1. Warm-up (5 min): Ask pupils what they'd like to know about the class: travel time, favourite subject, number of brothers and sisters. Choose one question together and write it at the top of the board.
2. Plan (10 min): With the class, decide the columns the question needs, the unit of each (minutes, a code) and the codes (W walk, B bus, C bicycle). Cross out anything the question doesn't need, such as names.
3. Collect (12 min): Pupils answer one by one while two tally-keepers make tally marks on the board, in groups of five. Everyone checks that the total matches the number of pupils present.
4. Type (8 min): If you have a phone or computer, type the counts into a spreadsheet with a pupil and save it as CSV. Without one, copy the tally into a tidy table in exercise books.
5. Wrap-up (5 min): Answer the question in one sentence from the table. Ask what else one more column could tell us, and whether we would really need it.

Choose a question with no sensitive answers: travel time and favourite subject are fine; family income or health are not. Keep the tally sheets until the table has been checked.

Discussion

- Which data did we decide not to collect, and why?
- What could go wrong if everyone wrote the minutes their own way?
- Why save the table as CSV instead of taking a photo of the board?

Exit ticket

- Write one question about our school, and the two columns you would need to answer it.

Try it in class tomorrow

Ask your pupils one question, like how long it takes them to get to school. Agree on the columns, the units and the codes together, collect the answers with tally marks, then type them into a sheet and save it as CSV.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Readings came in as "12", "12 mm" and "a lot". What would have stopped that?**

◯ Measuring twice a day ◯ A column headed "Rain (mm)", numbers only

◯ A bigger sheet of paper

2 **You want to know how long your pupils take to get to school. Which columns does your sheet need? Choose all that apply.**

☐ How they travel (W, B, C, M) ☐ Pupil number ☐ Minutes to school

☐ Each pupil's home address ☐ Parents' phone numbers

3 **Put the steps of a good data collection in order.**

☐ Type the counts into a spreadsheet

☐ Write the question

☐ Save it as CSV and check it

☐ Collect the answers with tally marks

☐ Decide the columns, units and codes

4 **Complete the sentence about CSV files.**

A CSV file is plain text: one _____ per row, _____ between the values, and the column _____ on the first line.

Word bank: picture · names · line · colours · commas

5 **The same attendance takes up the same space whether it's saved as CSV or as JSON.**

◯ True ◯ False

Answers · for the teacher

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1 **A column headed “Rain (mm)”, numbers only**

Plan the sheet before collecting: the unit goes in the column heading, “Rain (mm)”, and every cell holds just a number.

2 **Pupil number; How they travel (W, B, C, M); Minutes to school**

The question needs a row for each pupil (a number is enough), how they travel and the minutes.

3 **Top to bottom: 4, 1, 5, 3, 2**

The question comes first, because it decides what to collect.

4 **A CSV file is plain text: one line per row, commas between the values, and the column names on the first line.**

CSV stands for comma-separated values: each row is one line of text, the values are separated by commas, and the first line names the columns.

5 **False**

JSON repeats every column name in every record, so the same facts take more than twice as much space as in CSV.

For you to think about

Think of data your school collects. Which question does it answer, and is any of it collected “just in case”?